

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

May 2014

EE401 Electrical Machine Design-I

Date: 19.05.2014, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are the disadvantages of selecting higher specific electric loading and higher specific magnetic loading? [07]
- Q - 2 (a) Draw the winding diagram in the developed form for a 4 pole, 36 slots, 3 phase mush connected armature. [10]
- (b) Define the following terms: [04]
(a) Back pitch (b) Front pitch (c) Winding pitch (d) Commutator pitch

OR

- Q - 2 (a) What is the purpose of providing a dummy coil? [03]
- (b) In which situation the equalizer rings need to be provided in dc armature? [04]
- (c) 50kW, 4 pole, 600 rpm, dc shunt generator, the full load terminal voltage is 220V. Maximum gap density is 0.83 wb/m^2 and the armature ampere conductors per metre are 30000. Calculate suitable dimensions of armature core to give a square pole face. Assume that the full load armature voltage drop is 3% of the rated terminal voltage and field current is 1% of rated full load current. Ratio of pole arc to pole pitch is 0.67. [07]
- Q - 3 (a) Which are the factors to be considered in selecting the length of air gap? [07]
- (b) Determine the main dimensions, number of poles and the length of air gap of a 600kW, 500V, 900 rpm generator. Assume $B_{av} = 0.6 \text{ wb/m}^2$, $a_c = 35000$ per meter, $\Psi = 0.75$ and efficiency = 91%. Now the design constraints are: Peripheral speed should not be more than 40 m/s, frequency of flux reversal should not be more than 50 Hz, current per brush arm should not be more than 400A and armature mmf per pole should not be more than 7500A. The mmf required for the air gap is 50 % of armature mmf and $k_g = 1.15$. [07]

OR

- Q - 3 (a) Which are the methods for reducing the armature reaction effects? [07]
- (b) 12 pole, 500kW, 550V generator has a simplex lap winding consisting of 2484 conductors. 621 commutator segments. Ratio of pole arc to pole pitch is 0.7. (a) Calculate the demagnetizing mmf per pole at rated full load current if the brushes are shifted through three segments from GNA. What is the cross magnetizing mmf per pole? (b) Calculate the number of conductors that must be provided in each pole face if compensating winding is used for the machine. [07]

SECTION – II

- Q - 4 (a) Give the classification of insulating material as per IS 1271-1958. Give examples of each class. [07]
- Q - 5 (a) A 500 kVA transformer has a total loss of 7.5 kW at full load. The rate of heat dissipation from tank walls is 300 W/degree C rise and the heat energy required to raise its temperature by 1 degree C is 0.45 kWh, calculate (i) the final steady temperature rise and thermal time constant of transformer, (ii) half hour rating of the transformer to give the same temperature rise as in (i) if the copper loss at full load is twice the iron loss. [07]
- (b) For a transformer show that e.m.f. per turn E_t is given by $E_t = k\sqrt{kVA}$, where kVA = rating of transformer. What are the factors on which the value of K depends? Or What are factors that influence K and how? [07]

OR

- Q - 5 (a) Derive an equation for optimum design of transformer for minimum cost and minimum loss? [07]
- (b) Write a short note on "Change of parameters of a transformer with change of frequency". [07]
- Q - 6 (a) Calculate approximate overall dimensions for a 200 kVA, 6600/440 V, 50 Hz three phase core type transformer. The following data may be assumed: Emf per turn = 10V, maximum flux density = 1.3 Wb/m², current density = 2.5 A/mm², window space factor = 0.3, overall height = overall width, stacking factor = 0.9. Use a three stepped core. [07]
- (b) An 11kV, 25 Hz transformer has I²R, hysteresis and eddy current losses 1.6, 0.6 and 0.4 % of the output. What will be the % losses if the transformer is connected to 12 kV, 50 Hz supply assuming the full load current to remain the same? [07]

OR

- Q - 6 (a) Describe the constructional detail of bar and wound type current transformer. [07]
- (b) A 250 V, dc shunt motor has armature and field resistances of 0.24 ohm and 250 ohm respectively, and takes 4 A at no load. On full load the total input current is 70 A and the final steady temperature rise is 55 degree C. Estimate the overload that can be permitted for 1 hour period, if any such overload period is preceded by a period of at least 2 hours during which the motor is shut down. The temperature rise must never exceed 55 degree C. Take heating time constant as 1 hour and cooling time constant 1.2 hours. [07]
